(Please write your Enrollment Number) Enrollment No. 01601172021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:B.tech, Semester:VI
(May, 2024)

Subject Code: BAI 306	**Subject: Digital Image Processing**
Time :3 Hours	**Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 **(2.5*8=20)**

(a) Explain all types of adjacency involved in digital image processing with examples.

(b) Explain key issues involved in object recognition.

(c) Explain how power law and log transformation of an image is performed giving formula.

(d) What are the problems of Roberts Cross first order derivative and how it can be solved?

(e) Why impulse noise is known as salt and pepper noise?

(f) List 4 types of thresholding in segmentation and explain them.

(g) Explain edge detection in segmentation.

(h) Explain the 3 conditions which are checked while using structuring element in morphological operations.

UNIT-I

Q2 Perform histogram matching using 2 histograms given below: ***(10)***

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	100	80	90	30	50	20	60	0

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	0	0	0	80	100	60	70	90

Q3 Consider the following image segment: **(10)**

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

(a) Perform digital negative operation
(b) Perform contrast stretching with r1 =3, r2 = 5, s1 = 2 and s2 = 6
(c) Perform clipping operation with r1 = 2 and r2 = 5
(d) Perform bit plane slicing

UNIT-II

Q4 **(10)**

(a) Check whether DFT matrix is unitary or not
(b) Compute the 2D DFT of the 4 x 4 grayscale image given below

1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

(c) Compute inverse 2D DFT of the transform coefficients given by

16	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

Q5 Convert the given image using fourier transform and perform to smoothen image. Choose D0 = 0.5. Show step by step procedure. **(10)**

0	1	0	1
0	1	0	1
0	1	0	1
0	1	0	1

UNIT-III

Q6 (a) Calculate the efficiency of Huffman code for the following symbol whose probability of occurrence is given below: **(10)**

Symbol	Probability
a1	0.9
A2	0.06
a3	0.02
a4	0.02

(b) For the image given below, compute the degree of compression that can be achieved using Huffman coding of pixel values.

3	3	3	2
2	3	3	3
3	2	2	2
2	1	1	0

(c) For the same image given above, compute the degree of compression that can be achieved using run length coding, assuming 2 bits to represent the pixel value and 2 bits to represent the run length.

Q7 Apply region splitting and merging with threshold value as 3 for the given image. Also construct Quad tree structure for splitting **(10)**

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

UNIT-IV

Q8 A binary image X and structuring element B are given. Calculate: **(10)**

(a) X^c

(b) $Y1 = X \oplus B$

(c) $Y2$ = Erosion of X^c and B

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(d) Y3 = Erosion of X and B
(e) Y4 = $X^c \oplus B$
(f) What is relationship between X and B

X=

0	0	0	0	0	0
0	1	1	1	0	0
0	1	1	1	0	0
0	0	1	1	1	0
0	0	1	1	1	0
0	0	0	0	0	0

B:

0	1	0
1	1	1
0	1	0

Q9 (10)

(a) Verify $K = \begin{matrix} 37 & -15 \\ -15 & 37 \end{matrix}$ is a covariance matrix.

(b) Explain the steps of object recognition.

(c) Given an image calculate the 4 - chain code.

0	0	0	0	0	0
1	1	1	1	0	0
1	0	0	1	1	0
1	1	0	0	1	0
0	1	1	1	1	0
0	0	0	1	1	0